Multi-Document RAG System

Technical Report & Evaluation Framework

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1. Introduction & Overview

In September 2026 [cite: topic-3], a comprehensive multi-document Retrieval-Augmented Generation (RAG) system was designed and developed to address the challenges of extracting precise information from large corpora of unstructured documents. Traditional keyword search often falls short in understanding semantic intent, whereas LLMs alone suffer from hallucinations and lack access to private or domain-specific knowledge bases. This project bridges that gap by implementing a robust pipeline featuring automated text chunking, dense vector indexing, and rigorous evaluation via the RAGAS framework [cite: topic-3].

2. System Architecture

The system architecture is structured into modular layers designed to ensure scalability, low latency, and high retrieval accuracy:

- **Ingestion & Parsing Module:** Ingests documents across various formats (PDF, TXT, JSON), extracting clean text and stripping unnecessary artifacts.
- **Semantic Chunking Engine:** Splits documents into overlapping segments to preserve contextual continuity across chunk boundaries.
- **Embedding Generation:** Leverages sentence-transformers models (such as `all-MiniLM-L6-v2`) to convert text chunks into high-dimensional vector embeddings.
- **Vector Database Storage:** Utilizes ChromaDB for persistent vector storage and sub-millisecond similarity search using cosine distance metrics.
- **Generation & Synthesis:** Combines retrieved context windows with prompt templates to generate precise, grounded answers.

3. Evaluation via RAGAS Framework

To ensure system reliability and production readiness, the pipeline was evaluated using the RAGAS (Retrieval Augmented Generation Assessment) framework [cite: topic-3]. Key evaluation metrics include:

- **Faithfulness:** Quantifies whether the generated response is strictly derived from the retrieved context, ensuring zero hallucination.
- **Answer Relevance:** Evaluates how directly the output addresses the user's prompt without introducing extraneous details.
- **Context Precision:** Measures whether the most pertinent chunks are prioritized at the top of the retrieval list.
- **Context Recall:** Assesses whether all necessary information required to fully answer the prompt was successfully retrieved.

4. Conclusion & Future Directions

The multi-document RAG system successfully delivers verifiable accuracy and robust semantic search capabilities. Future enhancements will incorporate hybrid search methodologies combining sparse keyword retrieval (BM25) with dense vector search, alongside neural reranking to further optimize precision.